these were not pathological (disease-related) infections but in fact underground partnerships: far from suffering, the trees seemed to benefit from better nutrition. He invented a new term for the partnership —“mycorrhiza,” from the Greek *mykes*, meaning fungus, and *rhiza*, meaning root.

Mycorrhizae in action

False truffles are an example of the fungal side of this partnership. Nineteenth-century Prussian botanists had found these fungi under spruce trees, and noticed that each tree root was drawn toward a truffle, and wrapped in a fungal husk. Although they did not know it, the botanists were witnessing a phenomenon that is vital to many ecosystems.

Fungi are typically nourished by a supply of organic matter, from which they extract food by external digestion. A deep layer of forest litter is perfect. They pour digestive chemicals onto their meal and absorb the soluble organic compounds produced through a network of microscopic filaments—hyphae—called a mycelium.

Plants rely on root hairs to absorb water and minerals, such as nitrates and phosphates. But there is a limit to how far plant roots can grow and therefore what quantity of nutrients the root hairs can absorb. The hyphae of mycorrhizae can cover a much wider area, absorbing a much greater amount of minerals. When the fungal hyphae attach to the plant roots, they extend the root system, causing extra nutrients to seep into the plant.

Albert Frank realized that this partnership worked both ways. It was a winning combination for both plant and fungus. In exchange for passing on a share of its minerals, the fungus receives sugar from the plant—made by photosynthesis in the leaves and transported to the roots via the plant’s sap. This boosts the nutrient supply that the fungus derives from dead organic matter.